Complete Guide to Coding Questions on Important Topics

This PDF contains a comprehensive list of coding questions across key topics in computer science. The questions are categorized by topic, followed by solutions, examples, and explanations to help you prepare for coding interviews and enhance problem-solving skills.

Arrays & Strings (Basic)

Question: Find the maximum subarray sum (Kadane's Algorithm)

Solution: Solution: Use Kadane's algorithm to find the maximum sum in O(n) time.

Question: Find the majority element in an array

Solution: Solution: Use Boyer-Moore Voting Algorithm.

Question: Reverse an array

Solution: Solution: Iterate through the array and reverse elements in place.

Question: Find the first non-repeating character in a string

Solution: Solution: Use a hash map to count character frequencies.

Dynamic Programming (Basic)

Question: Fibonacci sequence using dynamic programming

Solution: Solution: Use a bottom-up approach to calculate Fibonacci numbers.

Question: Longest increasing subsequence

Solution: Solution: Use dynamic programming with binary search.

Question: Knapsack problem (0/1)

Solution: Solution: Use dynamic programming to solve the problem.

Question: Coin change problem (minimum coins)

Solution: Solution: Use a bottom-up DP approach.

Question: Longest common subsequence

Solution: Solution: Use dynamic programming for comparing subsequences.

Graph Algorithms

Question: Depth First Search (DFS)

Solution: Solution: Traverse the graph using DFS starting from a node.

Question: Breadth First Search (BFS)

Solution: Solution: Traverse the graph using BFS starting from a node.

Question: Detect a cycle in an undirected graph

Solution: Solution: Use DFS to detect back edges.

Question: Find the shortest path in an unweighted graph

Solution: Solution: Use BFS for finding the shortest path.

Question: Dijkstra's algorithm

Solution: Solution: Use a priority queue and greedy approach for shortest paths in weighted graphs.

Advanced Data Structures

Question: Design a Trie (Prefix Tree)

Solution: Solution: Implement insert, search, and delete methods.

Question: Implement a Binary Search Tree (BST)

Solution: Solution: Implement insert, delete, and search methods.

Question: Design a Min-Heap

Solution: Solution: Implement insert, extract-min, and heapify operations.

Question: Implement a Hash Map

Solution: Solution: Use arrays and handle collisions with chaining or open addressing.

Question: Design a Priority Queue

Solution: Solution: Use a heap to implement priority queue operations efficiently.

Backtracking & Recursion

Question: Solve N-Queens problem

Solution: Solution: Use backtracking to try placing queens in each row.

Question: Solve Sudoku puzzle

Solution: Solution: Use backtracking to try placing numbers in empty cells.

Question: Generate all permutations of a string

Solution: Solution: Use recursion to generate permutations.

Question: Generate all subsets of a set

Solution: Solution: Use backtracking to generate subsets.

Question: Solve the Rat in a Maze problem

Solution: Solution: Use backtracking to find paths in the maze.